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Frontier language-model agents violate payoff-scale invariance in the prisoner's dilemma, and cooperate without reciprocity

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Large language models increasingly act as agents that negotiate and decide for their users, so whether they cooperate is a governance question. We ask how two everyday levers shape it: the size of what is at stake, and the language the game is posed in. The first is unusual: multiplying every payoff by a positive constant leaves preferences, best replies, equilibria and replicator orbits untouched, so its predicted effect is zero by theorem, not convention. Five frontier models played an iterated prisoner's dilemma across ten payoff scales spanning five orders of magnitude, crossed with prompt language and persona, over 200,000 decisions. Cooperation moves in every model, by up to 0.636 (95% CI 0.588, 0.686), and differently in each, so averaging over models reports a movement no model makes. Its size depends on the prompt language; the scale also gates the persona instruction, reversing its sign in two models of five where every printed payoff is at most 1. The effect is already present on the opening move, and what is played changes too. An evolutionary baseline predicts the opposite. Cooperation rates published for language models should be read, and governed, alongside their scale sensitivity.

1. Introduction

Large language models are increasingly deployed as *agents* that negotiate, coordinate and decide on behalf of their users, in recommendation systems, negotiation tools and multi-agent assistants [1–4]. In these settings behaviour emerges from strategic interaction rather than from isolated model outputs, and the agents meet *cooperation dilemmas* whose resolution bears directly on safety, coordination and AI governance [5–8]. A fast-growing literature measures how they behave when they do, with the instruments of behavioural economics and cognitive psychology [9–11] and with the classical games [12]. Equally consequential is how far that behaviour adapts to the costs and benefits at issue, since a deployed agent meets the same dilemma at many different magnitudes [3].

A growing body of work evaluates language models through game-theoretic lenses, and in particular through matrix and repeated games, reporting systematic departures from equilibrium play, persistent cooperative biases and sensitivity to contextual factors such as language and incentives [13–20]. The prisoner's dilemma is the workhorse of that literature, its strategy space mapped and its predictions sharp [21–26]. Within it the models turn out to be individual rather than uniform, differing in how readily they retaliate [27], how readily they forgive [28], how much weight they give the game rather than the story around it [29] and how far an instructed disposition carries into conditioned reciprocity [30]. These studies share two unremarked reporting conventions. Each summarises a model by a cooperation rate, or a few of them, measured at one payoff matrix; and each reports what the agents did rather than the decision rule that produced it.

Evaluating strategic behaviour at the level that alignment and governance require therefore asks for more than aggregate outcomes. Drawing on behavioural and evolutionary game theory [31], a behavioural intention can be conceptualised as an agent's *strategy*: a decision rule mapping an interaction history to the next action [32–35]. The canonical strategies of repeated games, Always Cooperate, Always Defect, Tit-for-Tat and Win-Stay–Lose-Shift [31, 36], supply an interpretable vocabulary for such intentions, and prior work has shown that they can be inferred from noisy behavioural trajectories by supervised learning [32, 33] and extended by probabilistic models to mixed or stochastic play [37]. The inference is not free. Separating the memory-one rules from observed play has been shown to need a long run of observations even where the resident rule is fixed by construction [38, 39], which is why the read-out used here is reported as a resemblance, with its provenance beside it (§2.4).

In parallel, what a language model does is not invariant across the language it is addressed in [13]. Strategic reasoning, risk sensitivity and cooperation patterns depend on linguistic framing, sometimes by margins comparable to the differences between models [29, 40, 41]. Whether the strategies agents play are themselves language-dependent, and whether that dependence introduces systematic bias into multi-agent settings, is largely open.

That dependence is one instance of a wider pattern: what is measured on a language model depends on things the measurement was not about. Accuracy moves with the order of the in-context examples [42], and with formatting choices that leave the meaning of the prompt intact, by margins wide enough to swamp the differences between the models being compared [43]. Instructions about who the model is are equally consequential: a persona changes what a model produces [44], shifts measured capability [45] and, conditioned on a demographic backstory, reproduces the response distribution of the matching human subgroup [46]. What none of these manipulations can supply is a null.

The choice of units is not usually stated as a design decision at all, because in the theory it is not one. A payoff matrix is a von Neumann–Morgenstern utility, defined only up to a positive affine transformation [47, 48], so writing the same dilemma with cells ten times larger changes no preference, no best reply, no equilibrium and no orbit of the replicator dynamics. Under the theory the two matrices are not two games written differently. They are one game. The theory of the iterated dilemma inherits the indifference, turning on the ordering of the four payoffs and on ratios among them and never on their absolute size [21, 23, 24, 26], and the modelling literature normalises the scale away [49–51]. Current theory works at the indifference rather than merely permitting it: where social learners update on the payoff they last received, the condition for conditional cooperation to persist under strong selection loses the benefit and the cost of cooperation altogether and turns on the ordering of the payoffs alone [52].

The invariance belongs to the theory, not to people: human valuation is neither linear in magnitude nor anchored at zero [53], human choice reverses between logically equivalent descriptions of one problem [54], and human cooperation in the dilemma itself responds to the size of the stakes [55, 56]. That gap has been argued to be structural rather than incidental, with the linguistic description of an action entering the utility function directly and a shift proposed from outcome-based to language-based preferences [57].

Two things are established, and they have not been put together. The first is that what a language model does depends on features of the prompt that carry no task content: the order of the examples, the formatting, the persona, the language. The second is that multiplying every payoff by a positive constant carries no task content in the strictest sense available, because the matrix it produces is not a second game but the same one. What is not known is whether the strategic behaviour of a language model respects that invariance. To our knowledge, no study has asked. Every manipulation in the sensitivity literature could legitimately change behaviour, so a measured change under any of them is at worst a nuisance and at best a finding, against an expected effect nobody had specified. A rescaling is the one manipulation whose expected effect is specified, and specified as zero, so a measured departure has a stated reference value rather than a comparison condition.

What follows if the invariance fails is a question about what a published cooperation rate means. Machine behaviour proposes that autonomous systems be studied empirically, with the instruments of the behavioural sciences, because what such a system does is not deducible from how it was built [58]. Those instruments arrive with a known difficulty: a behavioural manipulation almost never carries a predicted effect, so a measured change has to be read against a comparison condition the experimenter chose. The invariance classes of decision theory are the exception, supplying a manipulation whose predicted effect is a stated number, and the number is zero. The consequences run through the strands of machine behaviour this study touches. The persona line is the one place where the experimenter addresses the agent about its social role, and whether it takes hold turns out to depend on the units the matrix was printed in. The departure is already present on the opening move, which locates it in the reading of the matrix rather than in the weighing of a history. And models of mixed human-machine populations carry the cooperativeness of the artificial agent as a parameter and ask where the population settles [59–63]; a rate partly determined by the experimenter's units is not the quantity those models need.

Together, these considerations let us put four questions to the rescaling:

(1) Does strategic behaviour survive multiplying every payoff by a positive constant, and by how much does it fail to?
(2) Does the answer depend on the prompt language, and on the persona the model is told to adopt?
(3) Does the rescaling change only how often a model cooperates, or also the strategy it is playing?
(4) Does the effect enter at the opening move, or through the course of play?

We answer them on a complete and exactly balanced grid of 10,000 dyads (§2), and read the answers against an evolutionary baseline computed on the same grid.

Here, we turn a theorem into an instrument. The contributions of the present paper are: **(i)** a payoff-rescaling protocol whose predicted behavioural effect is exactly zero, used to ask whether five frontier language models are playing the game they were shown (§2.2); all five depart from the invariance, by up to 0.636 in cooperation. **(ii)** a per-model analysis showing that the departure has a different shape in each model, to the point of anticorrelation, so that what a pooled analysis reports is a movement no model in the pool makes: the interaction of the payoff scale with the model carries 17.9% of the variance across cells against 5.8% for the main effect, and averaging the five curves gives a range of 0.154 where the mean within-model range is 0.249. The main effect is significant here only because one model departs far enough to survive the averaging, which is a fact about that model rather than a property of language models in general, and a panel assembled without it returns no significant main effect at all. **(iii)** evidence that the departure is structural and not a shift in level: the payoff scale gates the persona instruction, reversing its sign in two models of five and abolishing it in a third, play becomes measurably less consistent with any canonical memory-one rule in four models of five, and the effect is already present on the opening move, before any history exists. **(iv)** an analytical evolutionary baseline, recomputed on the same ten-value grid under matched payoffs and execution noise, which moves in the opposite direction to the corpus and so measures the size of the gap between what the game rewards and what these agents do (§3.7, appendix C).

An earlier and smaller collection on this project has been reported separately by the present authors [64]: three models at three payoff scales, in the same five prompt languages, with cooperation read off the trajectories and reported as rising with the stakes. The present article differs in what it collects and in what it claims. It collects six models rather than three and ten payoff scales rather than three, spanning five orders of magnitude, on a grid whose completeness and balance are enforced by the analysis build rather than assumed. It frames the rescaling as an invariance whose predicted effect is a theorem rather than as a manipulation of the stakes whose effect was expected. And it reports every quantity per model, which is what shows the movement to be heterogeneous, with two of the five running downwards, and the pooled statement to be an artefact of averaging.

2. Material and methods

2.1 Framework

FAIRGAME, the Framework for AI Agents Bias Recognition using Game Theory [13], supplies the computational infrastructure for this study. It supports systematic and reproducible evaluation of language models through controlled game-theoretic experiments: an experimental condition is declared in a configuration file that fixes the payoff structure, the horizon, the model backend and the prompt language; at run time the framework combines that declaration with language-specific prompt templates, simulates the repeated normal-form game, and logs the round-by-round trajectory. We extend it in three ways. First, with a payoff-scaling factor applied to the prisoner's dilemma, which is the manipulation this paper is about (§2.2). Second, with a completeness guard that refuses to release an analysis table unless the grid is exactly balanced and each game's recorded payoffs recover the scale its cell was run at. Third, with a strategy read-out that reports which canonical memory-one rule a trajectory most resembles, and with what provenance (§2.4).

The design is a complete factorial crossing of six language models, ten payoff scales, five prompt languages, four persona pairings and ten replicates. The unit of collection is the dyad, one ten-round game between two agents of the same model, contributing two agent-games, an agent-game being the ten decisions one agent takes in one game. Each of the 300 model-by-language-by-scale cells holds exactly 40 dyads, the four persona pairings by ten replicates; pooling the five languages gives 200 dyads and 400 agent-games in each of the 60 model-by-scale cells. The corpus is 12,000 dyads, 24,000 agent-games and 240,000 decisions. Five of the six models are reported in the main text and the sixth, Gemini 3.1 Flash-Lite, in the appendix, for the reason given with the models below. The five carry 250 of the cells, 10,000 dyads, 20,000 agent-games and 200,000 decisions, and are the grid every main-text figure and table is computed on. Balance is enforced rather than assumed (appendix A.1).

2.2 The payoff-scaled prisoner's dilemma and its invariance

Both agents receive the prisoner's-dilemma prompt of the FAIRGAME protocol [13] in its no-communication, announced-horizon condition. It casts the game as a criminal interrogation, states the four payoffs as penalties, names the two actions Option A and Option B, carries the running history of play, announces the ten-round horizon and instructs the agent to minimise its penalty. The row player's baseline payoff matrix is $\big((A,A)\mapsto(6,6),\ (A,B)\mapsto(0,10),\ (B,A)\mapsto(10,0),\ (B,B)\mapsto(2,2)\big)$. At $\lambda=1$ the agent is therefore shown $T=0$, $R=2$, $P=6$, $S=10$ in penalty units, in the usual temptation, reward, punishment and sucker notation. The ordering $T<R<P<S$ is the image of the familiar $T>R>P>S$ on utilities $u=-$penalty, and $2R>T+S$ holds on those utilities, so alternation is not preferred to mutual cooperation. Option A pays less in both columns and is therefore the dominant, defecting action, while mutual Option B at 2 beats mutual Option A at 6 and is the cooperative one; cooperation means Option B throughout this paper. The polarity was checked against the data as well as against the prompt (appendix A.1).

The manipulation multiplies all four cells by a positive constant λ (figure 1b), taking ten values spanning five orders of magnitude, $\lambda\in\{0.01,0.1,0.25,0.5,1,2,5,10,100,1000\}$, and changes nothing else in the prompt. Two details bear on interpretation: $T=0$ prints as 0 at every scale, so only three of the four numbers move, and the history block records realised penalties, so λ is present in every prompt of the game and not only the first. We treat the printed penalties as the agent's utilities, as the prompt's minimisation instruction directs; because λu with $\lambda>0$ is a positive affine transformation of a von Neumann–Morgenstern utility [47, 48], the predicted effect on an agent playing the game is exactly zero. It is this that separates the present manipulation from a manipulation of the stakes, whose behavioural effect is expected rather than specified [53, 54, 65]. An agent that instead applies a non-linear valuation to the printed penalties is not bound by the invariance, and §4 returns to readings of that kind. One boundary in the grid is a fact about the rendering rather than a free parameter: the largest number shown is 10λ, so every payoff printed is at most 1 exactly when $\lambda\leq0.1$, which two of the ten scales satisfy. The scaled matrix at all ten values is tabulated in appendix A.3.

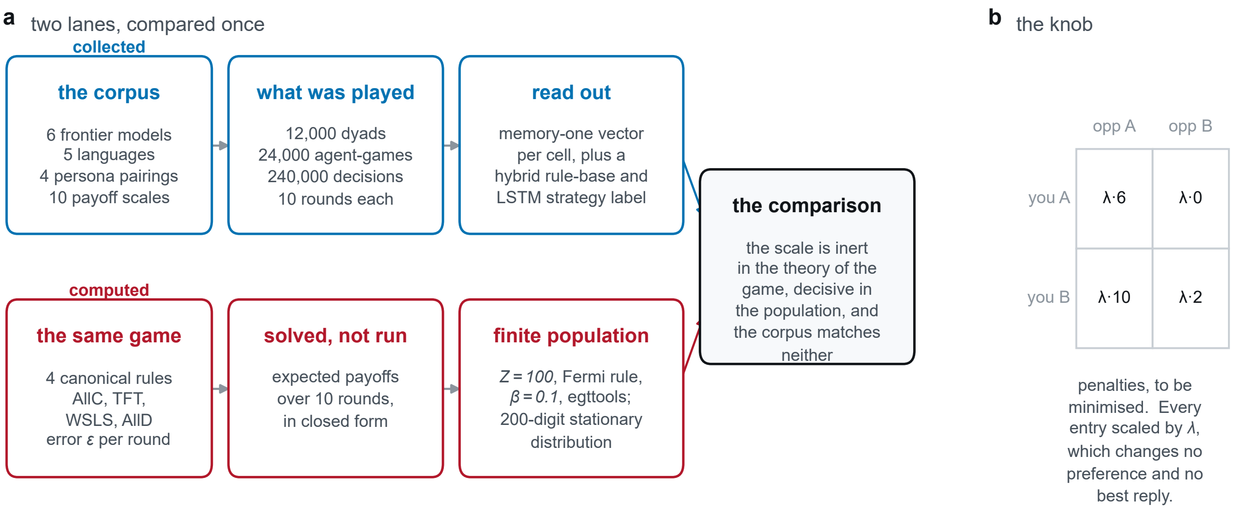


Figure 1. Overview of experimental design, payoff-scaling protocol and analysis lanes. (a) The empirical collection: six frontier language models played against themselves across ten payoff scales $\lambda\in[0.01,1000]$, five prompt languages, four persona pairings and ten replicates (12,000 dyads, 240,000 decisions). (b) The payoff-rescaling manipulation: multiplying the prisoner's dilemma penalty matrix by λ. Under classical decision theory and deterministic replicator dynamics, multiplying payoffs by a positive constant leaves preferences, best replies and equilibria invariant. (c) The analytical evolutionary lane: solving finite-population imitation dynamics ($Z=100$, Fermi selection $\beta=0.1$, execution noise $\epsilon=0.05$) at 200 significant digits across the same ten payoff scales, yielding the theoretical baseline against which the empirical corpus is compared.

2.3 Models, prompt languages, personas and protocol

Six frontier models were run, each against itself; two different models were never paired. They were Claude Haiku 4.5, GPT-5.4 Nano, Gemini 3.5 Flash-Lite, Qwen3 235B-A22B, Grok 4.20 and Gemini 3.1 Flash-Lite, served through the Kaggle Benchmarks model proxy on an OpenAI-compatible chat-completions endpoint at temperature 1.0. Their endpoint identifiers are listed in appendix A.2. The main text reports the first five. The sixth, Gemini 3.1 Flash-Lite, is reported in full in appendix B.7, and the reason is a property of its endpoint rather than of its results: its identifier is both undated and announced as provisional, so what was served under that name while the corpus was collected cannot be re-addressed afterwards. We state the cost of drawing the line there. The identifier of Gemini 3.5 Flash-Lite, which stays in the main text, is undated as well, so it carries a weaker form of the same risk; the line is between an identifier marked provisional and one that is not, and not between pinned and unpinned. Gemini 3.1 Flash-Lite is also the one model of the six whose strategy trend runs opposite to the rest (§3.5), which is a reason to report it prominently rather than to set it aside. Every quantity the main text gives for the five is given for all six in the appendix, which also states what including it changes, and on every conclusion of this paper that is nothing.

The prompt languages are English, French, Vietnamese, Chinese and Arabic, in templates byte-identical to the FAIRGAME resource files and checked against them before every collection run. Artefacts of those files are reproduced rather than repaired, the consequential one being that the Arabic and Chinese versions state the four cells as penalties but phrase the goal sentence as maximising a reward, so we treat the five languages as five distinct stimuli rather than five renderings of one (appendix A.4). Each agent also receives one line naming its persona, in the language of the prompt: in English, *You are cooperative.* or *You are selfish.* The two are crossed in all four ordered pairings, ten replicates each. The opponent's persona is never disclosed, so a pairing reaches a player only through realised behaviour and not before round 2, and we treat it as a design factor and a regression covariate rather than as information the agent held.

Every game runs ten rounds with simultaneous moves: both agents see the same history, rounds 1 to $r-1$, and the round is scored once both have replied. Sampling runs under a common-random-numbers scheme that gives a given round of a given replicate the same seed at every payoff scale. Decisions are elicited as free text and parsed by the protocol's own matcher. A reply it cannot resolve is regenerated, up to two further attempts, and if all three fail the decision is recorded as Option A, which under this framing is defection: the fallback can only push a measured cooperation rate down, never up, and it cannot manufacture the cooperative behaviour reported here. Each collection run asserts that its fallback rate is at most 0.02, but the achieved rates are not carried in the released per-game tables, so we cannot report how they vary with the payoff scale (appendix A.2).

2.4 Strategy read-out

While the framework provides complete gameplay trajectories and descriptive metrics such as cooperation rates, these capture what agents do rather than the decision rule behind it. Following prior work on inferring canonical strategies from noisy repeated-game data [32–34], we read every agent-game against the four canonical rules AllC, AllD, Tit-for-Tat and Win-Stay–Lose-Shift. The read-out reports which of the four a ten-round trajectory most *resembles*. It is not an identification of the strategy a model is running, and no claim in this paper treats it as one.

All four rules are memory-one, so exact matching is a lookup with no learning in it, and it returns a *set* rather than a label, because over ten rounds several rules routinely prescribe the same moves. The three provenances, the agent-games matched exactly by one rule (deduced), by several at once (ambiguous) and by none (unmatched), are recorded separately and never merged, and the same matcher yields a continuous quantity needing no labels at all, the number of rounds violating the nearest rule. Where the rules leave the answer open a single-layer LSTM [66] arbitrates. Trained on a synthetic corpus alone, which contains no language-model output, it reaches 98.5% accuracy in distribution and 93.6% and 81.1% on two execution-noise levels held out entirely (appendix A.6). It ranks only within the set the rule base returns, so it can never overturn a deduction; it decides alone only where no rule fits, and its share of the labels is very unevenly spread across the four rules, which bounds how the composition may be read (§3.5).

2.5 Statistical analysis

The cooperation rate of an agent-game is the share of its ten decisions that are Option B; a cell mean averages the agent-games of a cell, and round-level analyses use the binary decision itself. Because the two agents of a dyad play each other and the ten rounds inside an agent-game are strongly autocorrelated, every interval was computed by resampling whole dyads with replacement, 1,500 draws, percentile method [67], and every regression clustered its standard errors on the dyad. Because the range over the ten scales is a maximum minus a minimum, and so positive by construction, its bootstrap interval quantifies the precision of the range and not its distance from zero; the permutation test supplies the test. Because the null is a theorem, the matching test destroys the association with the payoff scale and leaves the rest of the design as

Table 1. Cooperation over the ten payoff scales, by model. Minimum, maximum and range, with a 95% confidence interval on the range from a bootstrap over whole dyads and a permutation test on the payoff-scale label (0.0005 is the floor of 2,000 draws). The last two columns test the nine scale contrasts jointly in a per-model logistic regression of the round-level decision, clustered on the dyad. Each entry pools 200 dyads per scale; ranges are computed before rounding. Full specification in the electronic supplementary material.

	cooperation					Wald test	
Model	min	max	range	95% CI	p_{perm}	χ^2_9	p
Claude Haiku 4.5	0.371	0.452	0.081	[0.062, 0.135]	0.0025	32.3	< 0.001
GPT-5.4 Nano	0.438	0.680	0.243	[0.200, 0.290]	0.0005	164.1	< 0.001
Gemini 3.5 Flash-Lite	0.682	0.872	0.190	[0.165, 0.240]	0.0005	129.0	< 0.001
Qwen3 235B-A22B	0.399	0.494	0.095	[0.068, 0.158]	0.0005	96.9	< 0.001
Grok 4.20	0.274	0.910	0.636	[0.588, 0.686]	0.0005	1227.3	< 0.001

observed: the scale label is shuffled among whole dyads, the unit to which the scale was assigned, and the range of the ten scale means recomputed, 2,000 draws, so 0.0005 is the smallest attainable p value.

For each model we fitted a logistic regression of the round-level decision on the payoff scale as a ten-level factor, with prompt language, own persona and opponent persona as covariates, $\lambda = 1$ as reference and standard errors clustered on the dyad, and tested the nine scale contrasts jointly by a Wald test (table 1). The variance decomposition is a type-II analysis of variance on the 250 cell means of the five models reported here, with terms for model, language and payoff scale and their three two-way interactions and no three-way term, so the residual on 144 degrees of freedom is the omitted three-way interaction; the same decomposition on all 300 cells is in appendix B.3. The persona result contrasts the two sub-unit scales with the remaining eight as a difference of differences with an interval from the same dyad bootstrap; where a trend across the whole grid is wanted instead, $\log_{10} \lambda$ enters as a continuous predictor. Analyses were run in Python; the package versions and the seeds are recorded with the archived pipeline in appendix E.

3. Results

Across the 240,000 recorded decisions of the whole corpus, the sixth model included, cooperation is 0.5895; the five models reported below carry 200,000 of those decisions.

3.1 Cooperation moves under a transformation that cannot matter

Cooperation depends on the payoff scale in all five models (figure 2a,b, table 1). The range across the ten scales runs from 0.081 (95% CI 0.062, 0.135) in Claude Haiku 4.5 to 0.636 (95% CI 0.588, 0.686) in Grok 4.20.

The observed range falls in the extreme upper tail of the permuted distribution in every model ($p = 0.0005$ in four of the five, the smallest value 2,000 draws can return, and $p = 0.0025$ in Claude Haiku 4.5), and the per-model logistic regression rejects the joint null on the nine scale contrasts ($\chi^2_9 = 32.3$ to 1227.3, all $p \leq 2 \times 10^{-4}$), the largest contrast against $\lambda = 1$ ranging from 0.36 to 3.74 in log-odds (figure 2c and table 1).

The shapes are unlike one another. Grok 4.20 cooperates 0.274 at $\lambda = 0.01$, the lowest rate these data contain, and 0.910 at $\lambda = 2$, the highest: the same model, shown the same game, is predominantly a defector at the bottom of the grid and predominantly a cooperator at the top. The others move less, and disagree with it and with each other. GPT-5.4 Nano rises steeply from 0.438 at $\lambda = 0.01$ and saturates near 0.65; Claude Haiku 4.5 dips to a minimum at $\lambda = 0.25$ and recovers; Gemini 3.5 Flash-Lite oscillates within a high band; Qwen3 235B-A22B drifts downward across the whole range.

3.2 The size of the effect depends on the prompt language

Neither prompt factor outside the game is neutral with respect to the payoff scale. Both the level of cooperation and its sensitivity to the scale differ with the prompt language (figure 3). Level differences are large and model-specific: Claude Haiku 4.5 cooperates 0.574 in Chinese and 0.234 in Vietnamese, while GPT-5.4 Nano cooperates 0.798 in English and 0.458 in Arabic.

One caveat attaches to the level comparison rather than to the scale result. The Arabic and Chinese templates, reproduced unaltered from the protocol, state the four cells as penalties while phrasing the goal sentence as maximising a reward (§2.3), so a level difference between two languages here is a difference between two stimuli and not between two renderings of one. It cannot produce a dependence on the payoff scale, because the wording is identical at every scale, and the ranges below are computed within a language.

The sensitivity differences are the new observation. The range of cooperation across the ten scales itself depends on the language: 0.130 in Vietnamese against 0.314 in French for Claude Haiku 4.5, and 0.162 in Chinese against 0.412 in English

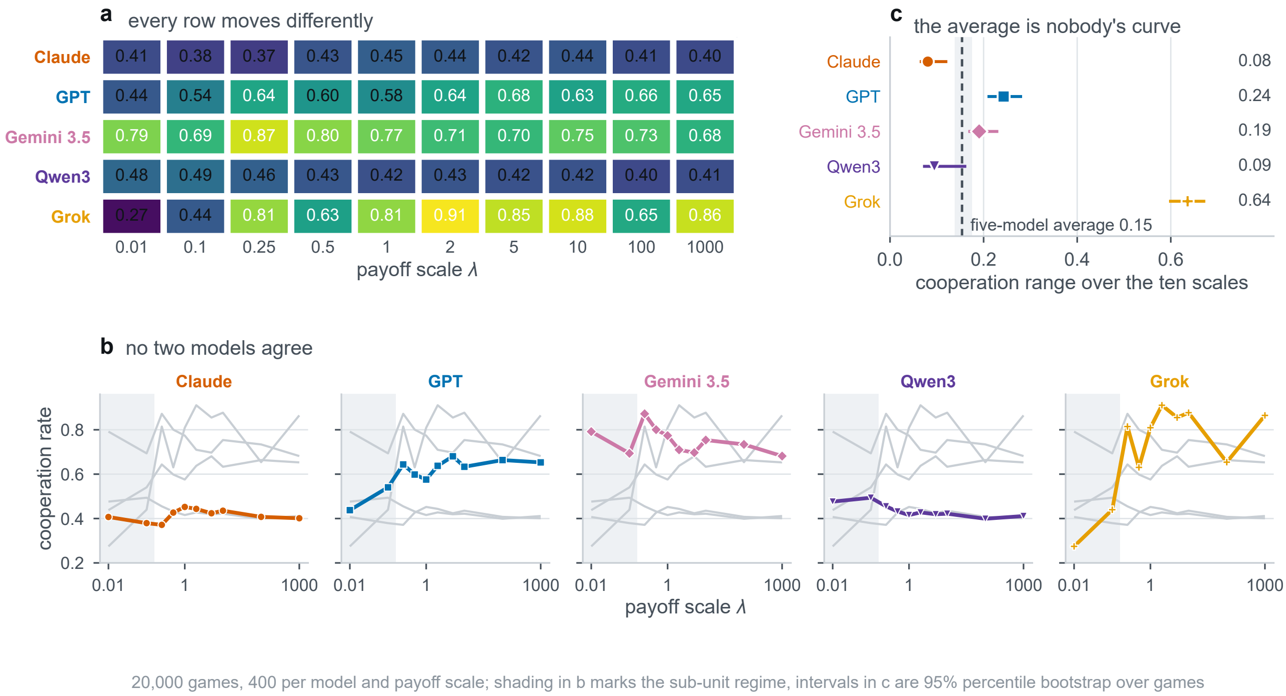


Figure 2. Cooperation against the payoff scale across five frontier models. (a) Model-by-scale cooperation matrix: mean cooperation for each of the five models at each of the ten payoff scales $\lambda \in [0.01, 1000]$, with the value printed in each tile. (b) Small multiples per model: the focal model is drawn with its designated colour and marker, with the other four models shown behind in grey for context. Shaded bands mark the sub-unit regime ($\lambda \leq 0.1$). (c) Within-model cooperation range (maximum minus minimum rate across the ten scales) compared against the pooled five-model range (0.154, vertical dashed line). Confidence intervals are 95% percentile intervals from 1,500 whole-dyad bootstrap resamples.

Table 2. Persona effect below and above the sub-unit boundary, by model. Cooperation under a cooperative persona minus cooperation under a selfish one, at the two scales where every printed payoff is at most 1 ($\lambda \leq 0.1$) and at the eight where some payoff exceeds it. Intervals are 95% confidence intervals from the dyad bootstrap. A sign flip needs both regime intervals to exclude zero on opposite sides, so a supra-unit interval straddling zero means the effect is abolished rather than reversed.

	persona effect, with 95% CI				
Model	$\lambda \leq 0.1$	$\lambda \geq 0.25$	shift	95% CI	sign flip
Claude Haiku 4.5	-0.222 $[-0.258, -0.186]$	$+0.207$ $[+0.189, +0.226]$	0.429	$[0.389, 0.471]$	yes
GPT-5.4 Nano	-0.135 $[-0.179, -0.091]$	$+0.018$ $[-0.001, +0.037]$	0.153	$[0.105, 0.200]$	no
Gemini 3.5 Flash-Lite	-0.456 $[-0.507, -0.406]$	$+0.224$ $[+0.207, +0.242]$	0.680	$[0.625, 0.736]$	yes
Qwen3 235B-A22B	-0.911 $[-0.931, -0.893]$	-0.679 $[-0.694, -0.664]$	0.233	$[0.210, 0.258]$	no
Grok 4.20	-0.441 $[-0.485, -0.396]$	-0.303 $[-0.326, -0.279]$	0.138	$[0.090, 0.185]$	no

for Gemini 3.5 Flash-Lite. Across the 25 model-language pairs it runs from 0.083, in Qwen3 235B-A22B in French, to 0.855, in Grok 4.20 in English (figure 3b). Scale sensitivity is a property of the model and the language together, not of the model alone.

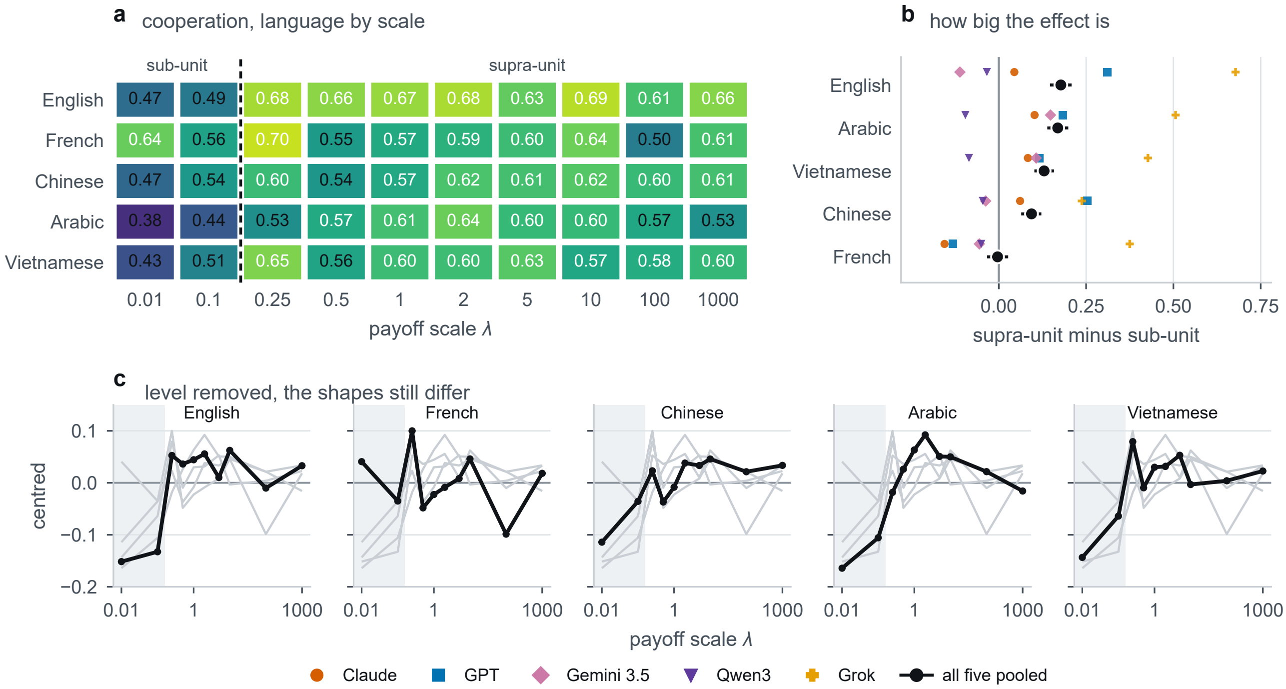
3.3 Heterogeneity, and what pooling does to it

That the shapes disagree has a direct consequence for reporting. Averaging the five model curves gives a pooled curve whose range is 0.154, while the mean within-model range is 0.249, an attenuation of 1.62 times (figure 2c). The pairwise correlations between model curves average -0.10 and run from -0.72 to $+0.85$, with six of the ten pairs negative (figure 2b). Curves that move in opposite directions cancel when added.

A variance decomposition on the 250 cell means makes the same point in another currency (table B.4, appendix B.3). The model accounts for 45.4% of the variance across cells and the model-by-language interaction for 17.0%. The payoff scale carries 5.8% as a main effect and is significant ($F_{9,144} = 10.64$, $p < 10^{-3}$), but its interaction with the model carries

Table 3. Rule agreement and rule distance against the payoff scale, by model. Left, the share of agent-games matched exactly by one canonical rule, below and above the sub-unit boundary, with the coefficient of a logistic regression of that indicator on $\log_{10}\lambda$; right, the mean number of rounds violating the nearest canonical rule at the smallest and largest scale, with the same trend by least squares. Both β are per decade of λ and their signs differ between models. Both cluster on the dyad, and neither block uses the learned classifier.

	exactly one rule fits (%)				rounds violating the nearest rule			
Model	$\lambda \leq 0.1$	$\lambda \geq 0.25$	β	p	λ=0.01	λ=10^3	β	p
Claude Haiku 4.5	17.9	3.5	−0.545	< 0.001	1.70	2.02	+0.073	< 0.001
GPT-5.4 Nano	9.8	7.6	−0.038	0.441	2.02	2.19	+0.038	0.052
Gemini 3.5 Flash-Lite	32.2	15.6	−0.242	< 0.001	0.48	1.36	+0.194	< 0.001
Qwen3 235B-A22B	72.1	53.6	−0.267	< 0.001	0.27	1.14	+0.202	< 0.001
Grok 4.20	44.6	17.1	−0.301	< 0.001	0.92	0.37	−0.118	< 0.001



20,000 games from five models, 400 per language and payoff scale; the interval on each pooled estimate is a 95% percentile bootstrap over games, and the small markers are the five models separately

Figure 3. Cross-lingual cooperation against the payoff scale. (a) Cooperation against the payoff scale λ, one panel per model, one curve per prompt language (English, French, Vietnamese, Chinese, Arabic). The shaded band marks the sub-unit regime ($\lambda \leq 0.1$). Annotations report the smallest and largest across-scale range among each model's five languages. (b) Across-scale range of cooperation for each model-by-language pair, showing that scale sensitivity varies by an order of magnitude across languages (from 0.083 in Qwen3 French to 0.855 in Grok English).

17.9%, three times as much ($p < 10^{-3}$), so most of what the payoff scale does is specific to the model rather than shared across models. Its interaction with the prompt language carries 3.6% ($p = 0.018$), the weakest test reported here.

That the main effect is significant at all is a fact about one model rather than about language models. It is carried by Grok 4.20, whose departure is large enough to survive being averaged with four others; a panel assembled without that model returns no significant main effect, with every remaining model still moving (appendix B.3). A pooled analysis therefore reports a quantity whose significance turns on which models happen to be in the panel, and whose size, 0.154 against a typical within-model 0.249, matches no member of it.

3.4 The payoff scale gates the persona instruction

The difference between the cooperation rates of the two persona instructions measures how much the persona takes hold. It depends on the payoff scale, and not gradually (figure 4, table 2). The natural boundary is where the printed numbers stop being sub-unit, at $\lambda \leq 0.1$ (§2.2). The persona effect shifts across it in all five models and every shift excludes zero, from 0.138 (95% CI 0.090, 0.185) in Grok 4.20 to 0.680 (95% CI 0.625, 0.736) in Gemini 3.5 Flash-Lite.

Whether the sign reverses is the stronger claim, and we hold it to a test on each regime rather than to a comparison of two point estimates. Two models of five reverse: in Gemini 3.5 Flash-Lite and Claude Haiku 4.5 the sub-unit interval lies wholly below zero and the supra-unit interval wholly above it, so below the boundary these models cooperate *more* when told they are selfish. In GPT-5.4 Nano the supra-unit interval straddles zero, so its effect is abolished at the boundary rather than reversed, and the other two models keep their sign throughout.

The boundary appears to be one of magnitude rather than of decimal notation. At $\lambda = 0.25$ the payoffs are $0, 0.5, 1.5, 2.5$: still decimals, but no longer all below 1, and there the persona effect already carries its supra-unit sign in both reversing models. It carries the supra-unit size as well only in Claude Haiku 4.5, where the effect at that scale is 0.206 against a supra-unit mean of 0.207; in Gemini 3.5 Flash-Lite it is 0.122, about half of that model's supra-unit mean of 0.224 and the smallest of its eight supra-unit values, so there the sign turns at the boundary while the magnitude keeps growing above it. The grid does not cross notation with magnitude fully, since every sub-unit scale in it is decimal-printed too.

Qwen3 235B-A22B and Grok 4.20 invert the persona instruction at every scale rather than only below the boundary (table 2), apparently resolving the story's ambiguity about who one cooperates with, one's partner or the authorities, opposite to the convention the payoff matrix encodes. Pooled over the grid, Qwen3 235B-A22B cooperates 0.072 under the cooperative persona and 0.798 under the selfish one. The inversion is not our subject, and we report it because excluding it would flatter the analysis; its magnitude depends on the payoff scale like everything else measured here (appendix B.4).

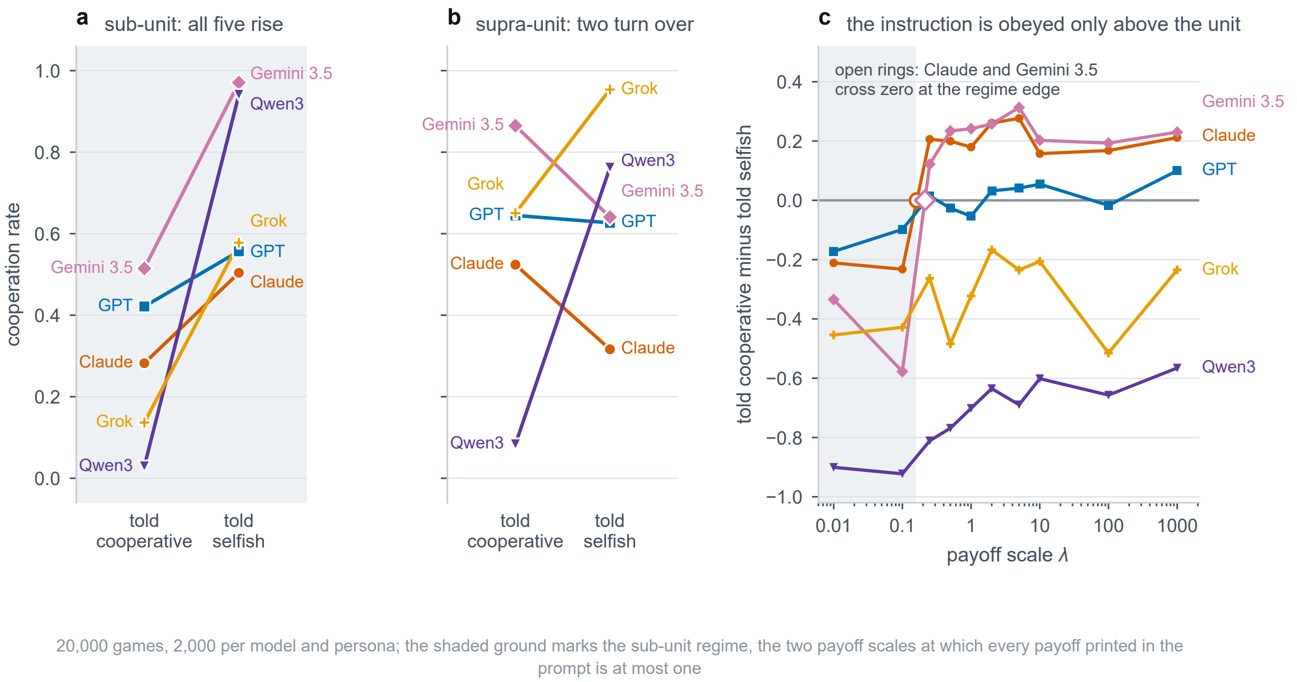


Figure 4. Persona effect and sub-unit inversion against the payoff scale. The persona effect is the cooperation rate under the cooperative persona minus that under the selfish persona. (a) Difference in cooperation rate across scales λ, with the sub-unit regime ($\lambda \leq 0.1$) shaded where all five models cooperate more under the selfish prompt. (b) Regime means comparing sub-unit ($\lambda \leq 0.1$, open circles) with supra-unit ($\lambda \geq 0.25$, filled circles) regimes (table 2).

3.5 The scale changes what is played, not only how often

Two models with the same cooperation rate can be playing different games, so we read each agent-game against the four canonical memory-one rules (§2.4). What the read-out returns is a resemblance and not an identification, and the labels below should be read that way throughout. Over the whole corpus, all six models included, 23.0% of agent-games are matched exactly by one rule, 17.5% by several at once and 59.6% by none.

Play becomes less rule-governed as the payoffs grow in four models of five, and shows no trend in the fifth (table 3, appendix B.5). Pooling the five, the share matched exactly by one canonical rule falls from 38.6% at $\lambda = 0.01$ to 17.3% at $\lambda = 10^3$, and the share matched by none rises from 51.6% to 64.0%. The trend is significant and negative in Claude Haiku 4.5 ($\beta = -0.545$ per decade of λ), Grok 4.20 (-0.301), Qwen3 235B-A22B (-0.267) and Gemini 3.5 Flash-Lite (-0.242), and absent in GPT-5.4 Nano ($p = 0.44$).

No model in the main text runs the other way on this measure, and one model in the corpus does, so the claim should not be read as uniform. The sixth model, Gemini 3.1 Flash-Lite, which is reported in the appendix because its identifier names an unpinned preview endpoint rather than because of anything in its results (§2.3), carries a significant positive trend ($\beta = +0.125$), its play becoming *more* consistent with a canonical rule as the payoffs grow. It is the only model of the six that does. Over the whole corpus the result is therefore four models moving one way, one showing no trend and one moving the other, and that is the form the evidence supports.

The continuous version needs no labels at all: the number of rounds violating the nearest canonical rule (table 3, appendix B.5). It rises from 0.48 to 1.36 in Gemini 3.5 Flash-Lite and from 0.27 to 1.14 in Qwen3 235B-A22B between the smallest and largest scale, with significant positive trends there and in Claude Haiku 4.5. Grok 4.20 moves the other way, falling from 0.92 to 0.37 ($\beta = -0.118$, $p < 0.001$): within the main five, three drift away from the canonical rules as the payoffs grow and one drifts towards them.

The composition shifts too, and the models do not shift together (figure 5). Pooled across the five, the AllC share rises from 38.1% to 45.8% of agent-games between the extreme scales and the Tit-for-Tat share from 9.1% to 12.7%, both at the expense of AllD, which falls from 43.6% to 28.3%. That pooled figure conceals the same cancellation: GPT-5.4 Nano moves from 29.8% to 50.7% AllC while its AllD share falls from 46.0% to 17.5%, whereas Gemini 3.5 Flash-Lite moves the other way on AllC, from 73.8% to 51.5%, with Tit-for-Tat roughly doubling from 6.8% to 14.2%, and Grok 4.20 inverts, its AllC share rising from 18.0% to 79.0% as AllD falls from 66.5% to 8.8%.

This last result leans hardest on the learned component, and the provenance is not uniform across the four labels: AllC and AllD are anchored in part by exact matching, whereas over the corpus 94.6% of the Tit-for-Tat labels and 97.1% of the Win-Stay–Lose-Shift labels are attributions made where no canonical rule fits (appendix B.5). A movement in the Tit-for-Tat share is therefore a statement about which canonical rule the play most resembles, not a claim that the model is running Tit-for-Tat.

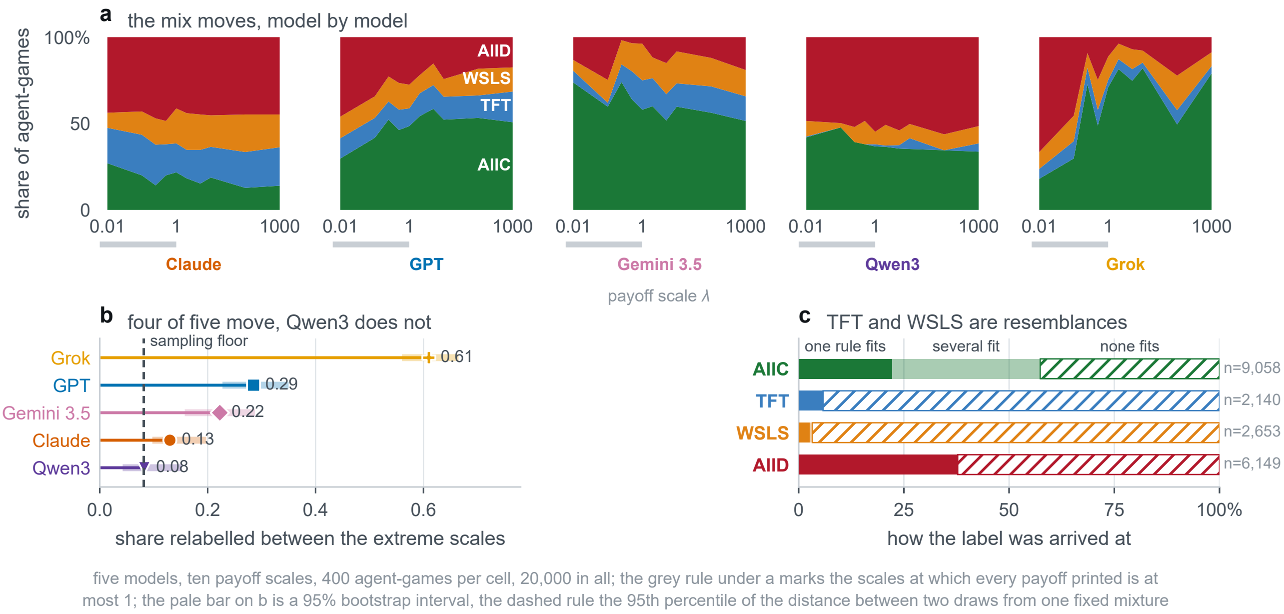


Figure 5. Strategy attribution across payoff scales and provenance breakdown. (a) Stacked area share of canonical memory-one strategy attributions (AllC, AllD, TFT, WSLS) across the ten payoff scales for each of the five models. (b) Strategy read-out provenance: exactly deduced (single canonical match), ambiguous (multiple matching rules), and unmatched (LSTM attribution on trajectories consistent with no textbook rule). Over the corpus, 94.2% of Tit-for-Tat and 96.8% of Win-Stay–Lose-Shift labels are attributions on unmatched play, establishing that labels indicate resemblance rather than faithful execution.

3.6 The effect is present before any strategic interaction

A rescaling could act through the course of play, by changing how a model reads its opponent's history, or on the reading of the matrix itself. The opening move separates the two: at round 1 nothing but the matrix and the instructions has been seen.

The effect is already present at round 1 (figure 6): in four models of five the range of opening-move cooperation across the ten scales exceeds the range over rounds 2 to 10, by factors of 2.0 in Claude Haiku 4.5, 1.9 in Gemini 3.5 Flash-Lite, 1.3 in GPT-5.4 Nano and 1.2 in Grok 4.20; the exception is Qwen3 235B-A22B at 0.47. The two windows rest on 400 and 3,600 decisions per cell, so their ranges are not variance-matched and these ratios are descriptive rather than tests.

The reading we take from this is deflationary about dynamics. Whatever the payoff scale does, most of it is already done before the game has begun. That is the timing an initial disposition read off the magnitudes of the matrix would produce. It does not rule out a further effect on how history is weighed, and this design does not separate the two.

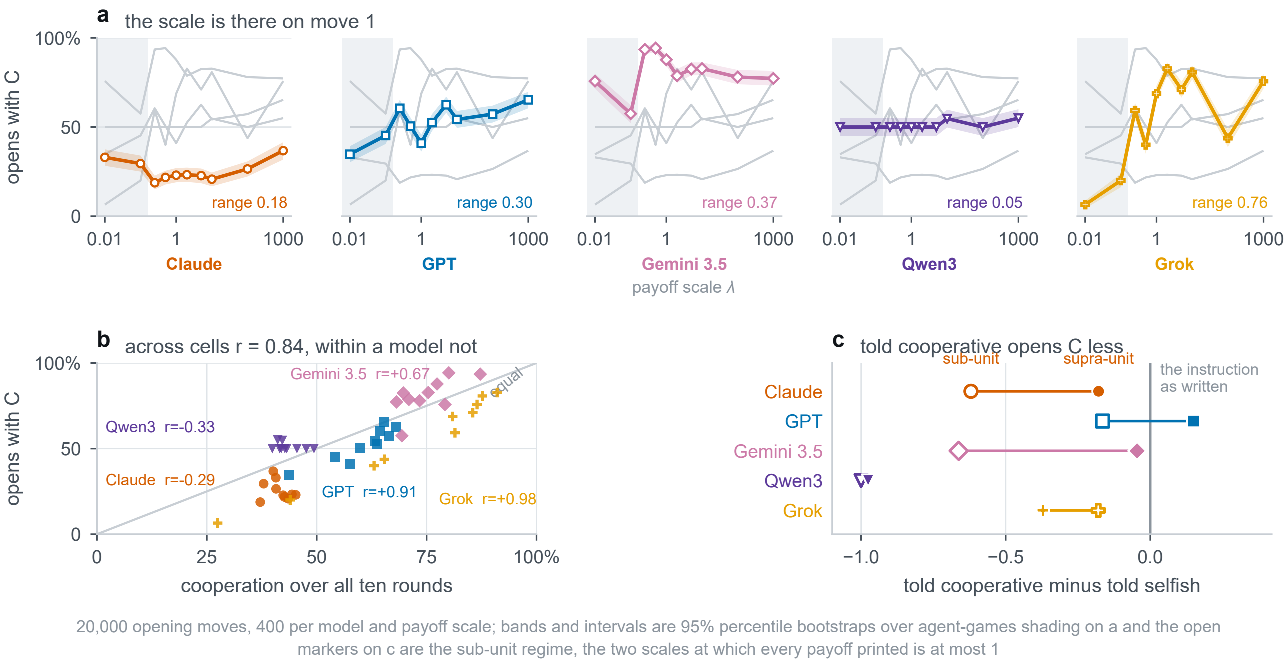


Figure 6. Opening move cooperation against the payoff scale. At round 1 the agent has observed only the matrix and instructions, with zero interaction history. (a) Round 1 cooperation rate across payoff scales λ for each model. (b) Across-scale cooperation range on the opening move (solid bars) compared to rounds 2–10 (hatched bars), demonstrating that scale sensitivity is established before any history exists.

3.7 Where the scale does act: the evolutionary baseline

The scale is inert in the theory of the game, but it is not inert everywhere, and locating the one place it does act sharpens what the corpus is being compared against.

It does nothing to the deterministic dynamics. Multiplying the payoff matrix by a positive constant multiplies the replicator field by that constant, so the orbits are the same curves traversed faster and every rest point stays where it was. Figure 7 draws the field on the four faces of the three-simplex the four canonical rules span, once, because there is only one picture to draw: the matrix at $\lambda = 1000$ is exactly 10^5 times the matrix at $\lambda = 0.01$, and over 2,000 random population states the normalised fields agree to 4×10^{-14}.

It does act on a finite population. The same matrix, sign-flipped so that lower penalties become higher utilities, drives a pairwise-comparison imitation process over the same four rules under a per-round execution error ($Z = 100$, $\beta = 0.1$, $r = 10$, $\epsilon = 0.05$; appendix C). There λ and the selection intensity β enter the Fermi update only through their product, so raising the scale is raising selection pressure, and the small-mutation-limit stationary distribution moves accordingly. Figure 8 shows what that does to the invasion structure: at $\lambda = 0.01$ every fixation probability is within a whisker of drift and the four rules share the population; at $\lambda = 1$ the familiar structure appears, with AllC invaded and AllD accumulating;

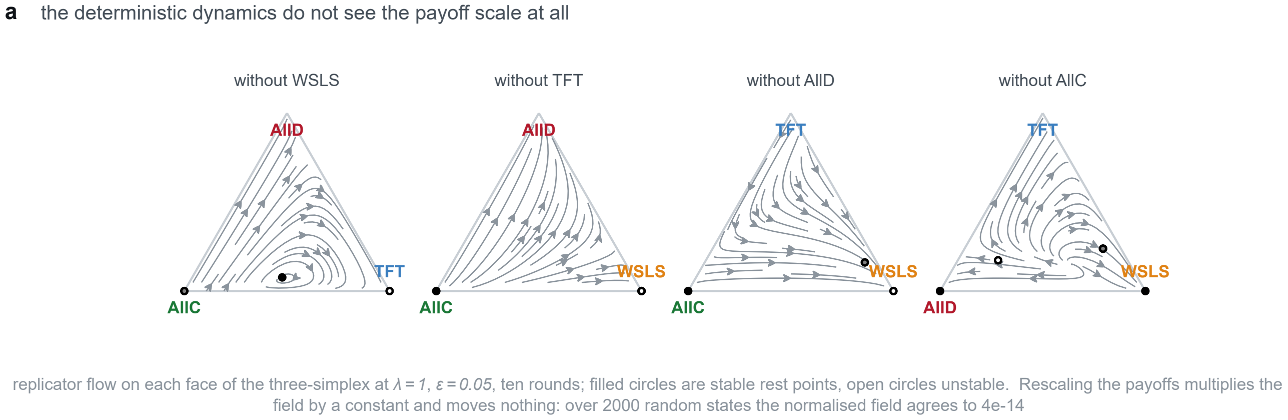


Figure 7. The deterministic dynamics do not see the payoff scale. Replicator flow on each of the four faces of the three-simplex spanned by the canonical rules, one face per strategy left out, at $\lambda = 1$, $\epsilon = 0.05$ and ten rounds. Filled circles are stable rest points, open circles unstable. Only one row is drawn because rescaling the payoffs multiplies the field by a constant and moves nothing: the matrix at $\lambda = 1000$ is exactly 10^5 times the matrix at $\lambda = 0.01$, and over 2,000 random population states the normalised fields agree to 4×10^{-14}.

by $\lambda = 10$ AllD holds 96.9%. Combining evolutionary modelling with language-model simulations in this way has proved informative in recent studies of governance dilemmas [40, 68].

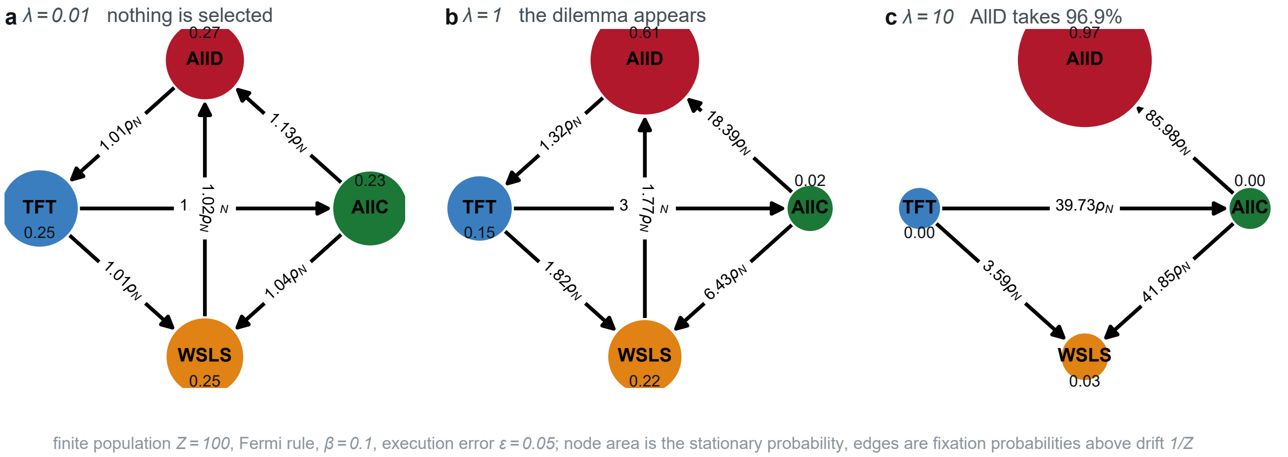


Figure 8. Where the payoff scale does act: the invasion structure of a finite population. Nodes are the four monomorphic states, node area is the small-mutation-limit stationary probability, and an edge is drawn where a single mutant fixes more often than neutral drift $1/Z$ would carry it, labelled in multiples of that drift. $Z = 100$, Fermi pairwise comparison, $\beta = 0.1$, $\epsilon = 0.05$. Because λ and β enter the update rule only through their product, raising the scale is raising selection pressure. The panels stop at $\lambda = 10$ because the float64 fixation probabilities underflow beyond it; those cells are solved at 200 digits instead (appendix C).

The baseline and the corpus then move in opposite directions (figure 9). The baseline's AllD share rises from 0.267 at $\lambda = 0.01$ to 1.000 by $\lambda = 100$. The corpus does the reverse and then flattens: its pooled AllD share falls from 43.6% to 28.3% while its AllC share rises from 38.1% to 45.8%, and at $\lambda = 1000$ the two accounts stand at 1.000 against 0.283.

That gap is not an artefact of the three constants the baseline was solved at. We solved the same model on 80 settings, $Z \in \{50, 100, 200, 500\}$ crossed with $\beta \in \{0.01, 0.05, 0.1, 0.5, 1\}$ and $\epsilon \in \{0, 0.05, 0.1, 0.2\}$, at the same ten scales, 800 cells each carried at 200 significant digits (figure 10). The AllD share rises with the payoff scale in 80 of 80 settings and passes 90% in 80 of 80; the median rise is 0.716 and the smallest is 0.195. The rise is monotone in 61 of the 80, and every one of the 19 exceptions is at $\epsilon = 0$, where the share dips before it rises, which is the qualification the baseline already carries (appendix C). The direction of the prediction is therefore a property of the model rather than of its parameters, and the empirical curve does not lie inside the family at any setting.

We read this as a measurement of the gap rather than as a falsification of either object. What the gap sizes is everything these agents bring that an evolutionary model does not, from alignment training to the cooperative bias of a human text corpus; the baseline is also language-blind by construction, whereas the model-by-language interaction carries 17.0% of the variance across cells here. Section 3.8 identifies one concrete component of it. A precision audit of the numerics and a comparison matched to execution-noise levels estimated from the read-out itself are in appendix C.

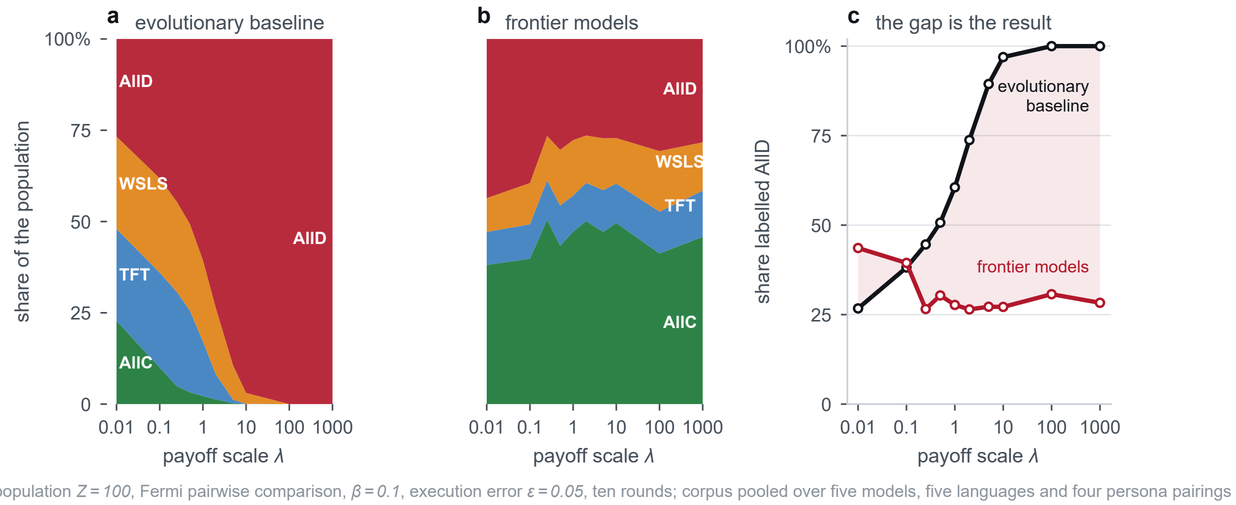


Figure 9. The evolutionary baseline and the corpus move in opposite directions. (a) The baseline's stationary distribution over the four canonical rules across the ten payoff scales, drawn as a stacked band. (b) The corpus's strategy mix on the same axis, drawn identically. (c) The AllD share alone, both accounts on one axis, with the gap shaded. Baseline at $Z = 100$, $\beta = 0.1$, $\epsilon = 0.05$, ten rounds; the corpus pooled over five models, five languages and four persona pairings.

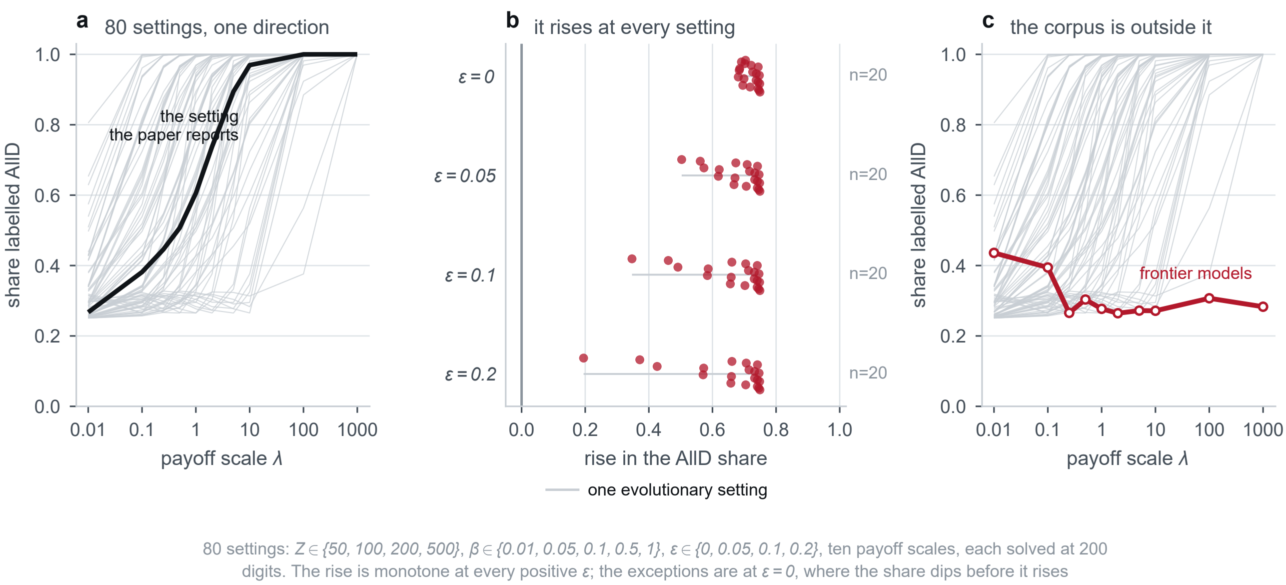


Figure 10. The baseline's direction is a property of the model, not of its parameters. (a) The AllD share against the payoff scale, one hairline per evolutionary setting, with the setting reported in the text drawn heavy. (b) The rise in the AllD share from the smallest scale to the largest, by execution error. (c) The corpus on the same axis as the family. Eighty settings, $Z \in \{50, 100, 200, 500\}$ crossed with $\beta \in \{0.01, 0.05, 0.1, 0.5, 1\}$ and $\epsilon \in \{0, 0.05, 0.1, 0.2\}$, ten payoff scales, 800 cells each solved at 200 significant digits. The rise is monotone at every positive ϵ; all 19 exceptions are at $\epsilon = 0$.

3.8 What the agents condition on

The baseline rests on a mechanism, and the mechanism can be measured directly in the corpus rather than inferred from the gap. What sustains cooperation against invasion by defectors in the evolutionary account is *reciprocity*: conditioning one's next move on what the partner just did. Tit-for-tat is the pure case.

The predecessor of a round is a pair, this agent's own last action and its partner's, so the four conditional cooperation probabilities form a two-by-two design and decompose without remainder into a level and three contrasts. We name them for what they are. *Reciprocity* is the partner main effect, $p(C \mid \text{partner } C) - p(C \mid \text{partner } D)$, and is what tit-for-tat is made of. *Persistence* is the own main effect, $p(C \mid \text{own } C) - p(C \mid \text{own } D)$. *Matching* is the interaction, and is what win-stay lose-shift is made of. The four canonical rules sit at known coordinates, which is what makes the space readable: AllC and AllD are both zero on all three contrasts and differ only in level, tit-for-tat is reciprocity 1, win-stay lose-shift is matching 1.

Written in those terms the corpus is not off the reciprocity axis by a little (figure 11). Averaged over the 212 model-by-scale-by-language cells with all four predecessors observed at least thirty times, the own effect exceeds the partner effect in every one of the five models, by factors of 1.2 (Claude Haiku 4.5, 0.288 against 0.243), 2.0 (GPT-5.4 Nano, 0.277 against 0.141), 3.4 (Gemini 3.5 Flash-Lite, 0.573 against 0.169) and 3.7 (Grok 4.20, 0.690 against 0.189); in Qwen3 235B-A22B the partner effect is negative (-0.095) while the own effect is 0.749. A principal-component analysis of the raw five-vector, given no knowledge of the contrasts, recovers the same geometry.

These agents cooperate, and at the larger scales they cooperate a great deal, but they are not doing it by answering their partner. They are conditioning predominantly on their own previous action, which is the one thing an evolutionary account of cooperation gives no weight to, and it is a concrete component of the gap in section 3.7: a population of agents that ignores its partner cannot support the reciprocal equilibria the baseline is built out of, whatever its cooperation rate.

One caution governs the arithmetic. The contrast $p_{CD} - p_{DC}$, which is the tempting one-line summary because it compares the two predecessors on which the pair disagreed, is not a reciprocity measure: it equals persistence minus reciprocity exactly, so it confounds the two main effects and can report anti-reciprocity where the partner effect is in fact positive.

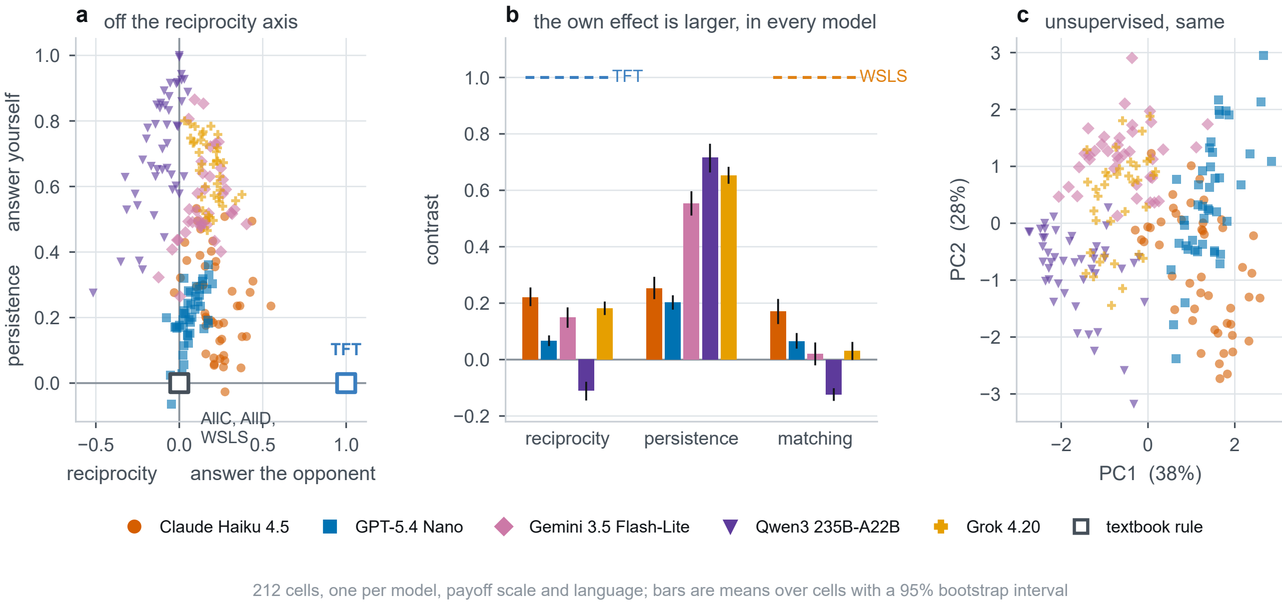


Figure 11. The agents condition on themselves, not on their partner. (a) Cells in the reciprocity by persistence plane, with the canonical rules at their exact coordinates; AllC, AllD and win-stay lose-shift all have both main effects zero and so coincide at the origin. (b) The three contrasts per model, against the value each canonical rule would give, with 95% bootstrap intervals over cells. (c) The first two principal components of the raw five-vector, which is given no knowledge of the contrasts and recovers the same geometry. A cell is one model, payoff scale and prompt language; a predecessor observed fewer than thirty times yields no estimate and the cell is dropped, leaving 212 of 250.

4. Discussion

4.1 Payoff-scale invariance is violated

The measurement reported here is unusual in having a null that is a theorem. That property lets a movement in cooperation be read as a departure from an invariance rather than as the difference between two conditions whose expected gap was never specified. The predicted difference between a matrix and the same matrix with larger cells is not small, or contested. It is zero.

All five models depart from it, and none is close to invariant. What makes the departures hard to set aside is not their size but their structure: they are present on the opening move, before any history exists, their size depends on the language the game is posed in, they gate the persona instruction at a boundary of magnitude, and they change what the transcripts look like as sequences. Sampling variability alone is difficult to reconcile with a pattern of that shape. The models appear to be responding to something in the matrix other than the game it encodes, though this design does not identify what.

4.2 Heterogeneity, and why pooled designs miss it

The pooling result deserves emphasis, because it is the mechanism by which an effect of this size could have gone unnoticed. The cause is arithmetic rather than statistical: the shapes disagree, and several pairs are anticorrelated. Heterogeneous effects that cancel are invisible to pooled designs by construction.

In this panel the cancellation is partial rather than total. A main effect of the payoff scale does survive the averaging, and it is significant, but it carries a third of the variance its interaction with the model carries, and the pooled range understates the typical model by a factor of 1.62: a statement about an average model that nobody deploys. A conclusion that changes when one model joins or leaves a panel is not a conclusion about language models. Our own pooled estimate would have been a null had one model been absent from the panel, and it is not a quantity we would defend for its own sake.

The pooled estimate is therefore not a weaker version of the per-model result but a different quantity, and it is not a question of power either: more dyads or more models only tighten the interval around an average of curves that point in different directions. The remedy is a per-model analysis, not a larger sample. The same logic runs one level deeper, since the effect depends on the prompt language as well as on the model, so a per-model number pooled over languages is the same trick played on a smaller stage. The heterogeneity is not a nuisance to be averaged away. It is the finding.

4.3 Implications for benchmarking and governance

A cooperation rate published for a language model is currently understood as a property of the model. On this evidence it is a joint property of the model, the units the matrix was written in, and the language the game was posed in. The units are the awkward member of that list, because they are the one nobody is asked to record: framings get described, personas get quoted, prompts get reproduced in appendices, while the payoff scale is fixed once, silently, on the authority of a theorem saying the choice is empty. Two laboratories running the same protocol on the same model, with matrices differing by a constant factor, could report cooperation rates differing by as much as 0.636.

We would suggest that a scale-sensitivity measurement accompany a published cooperation rate, in the way an error bar does, and be measured per model, most of all where such rates leave the benchmark and become the parameters of a population model. The same argument applies to safety auditing. An audit that runs in English at one fixed payoff matrix tests a single point on a much larger behavioural surface, and the two cheapest stress dimensions to add are the two measured here: a sweep over the payoff scale, and a parallel run in the prompt's non-English variants. Our central claim is complementary to a parallel line of work asking how external *mechanisms* change cooperation, through contracts and mediation [69], moral framing [70] or pre-play communication [71]. Payoff scale joins that list as an intervention on the numerical structure of the game rather than on the institutions around it. The behavioural profiles it produces are also the input that models of hybrid human-machine populations consume [7], and a profile that depends on the experimenter's units is not the quantity such a model needs.

4.4 Interpretation and possible mechanisms

Two observations constrain what an explanation must account for. The boundary gating the persona instruction appears to be one of magnitude rather than of notation, a separation only as strong as the single scale that supports it (§3.4); and the payoff scale moves how closely play tracks any canonical memory-one rule at all, and shifts the mix of labels with it. The scale is not turning a single dial marked cooperativeness; it is changing the shape of the trajectory.

Several accounts are consistent with both. The printed magnitudes may be read as stakes, which is exactly what a rescaling does not change; numerals of different magnitudes are tokenised differently and are met at very different frequencies in training text, so a representation that changes with the digit string could change what follows from it with no notion of stakes involved [72]; training text may associate large quantities with caution; and sub-unit payoffs may fall outside the range in which payoff-like numbers are ordinarily met. We can order none of these. The design measures departures from an invariance; it does not manipulate tokenisation, vary notation at fixed magnitude beyond the one contrast above, or move the framing that supplies the stakes. Nothing here should be read as identifying a mechanism.

There is a constructive reading as well as a cautionary one. A programme is under way to widen the utility function beyond the economic consequences of an action, so that the language describing the action enters it directly, with the replicator dynamics named as the destination [57]. Our result sits one level below that programme rather than beside it. The evidence for language-based preferences comes from manipulations that could legitimately have mattered, since a label carries connotation, whereas a positive rescaling carries none and is inert by theorem, and the behaviour still moves. A utility that mixes a payoff with a score computed from language is no longer scale-free in the payoff argument unless the two terms are rescaled together, so before such an object is carried into an evolutionary dynamic its invariance class has to be stated. What these data say is that at least one class of agents does not behave as though the payoff argument were scale-free, and that anything built to describe them will have to account for it.

4.5 Limitations and future work

This design does not identify the mechanism inside the model that produces the departures it measures across a broad but finite grid. Six models cannot support claims about model families or about parameter count, and what generalises here is that the shapes differ rather than what any one of them is. The corpus is one game, one prompt template family and one horizon, so whether the same curves appear in other dilemmas or in longer play is untested.

A sixth model, Gemini 3.1 Flash-Lite, was run on the identical grid and is reported in full in appendix B.7 rather than here, because its identifier names an unpinned preview endpoint and what was served under that name during collection cannot be recovered (§2.3). Two consequences are worth stating rather than leaving to be discovered. First, it is the only model of the six whose strategy trend is positive, so the four-of-five statement in §3.5 reads, over the whole corpus, as four models of six moving one way, one showing no trend and one moving the other; the second form is the one the corpus supports. Second, including it changes no conclusion here: the all-six pooled range, attenuation, shape correlations and variance decomposition are in appendix D.1, and every sign, significance and ordering in them agrees with the five-model versions reported above.

One feature of the stimuli bounds the language comparison. Two of the five templates state the payoffs as penalties while phrasing the goal as maximising a reward, and we reproduced rather than repaired them so that the corpus matches the published protocol. Because that wording is fixed across the ten scales it cannot generate a scale effect, and the within-language sensitivities stand; it does mean that a difference in level between two languages is not attributable to the language alone.

Our reference for calling a departure large is the sampling variability of these data, a bootstrap over dyads and a permutation null, rather than an independent repeat of the whole protocol. A measured test-retest floor, from sweeps re-run under identical conditions, is the obvious next measurement. Clustering on the dyad addresses the dependence between the two agents of a game and between the rounds within an agent-game, but it does not turn the design into a mixed-effects model with random intercepts for replicate and game, which would be the more complete treatment of a nested design.

The strategy read-out carries a further caveat, which is why its provenance is reported wherever a label appears (§2.4). A strategy is a decision rule and not a decision, so it is never observed directly, and even where the resident rule is fixed by construction, separating the memory-one rules from observed play has been shown to need a long run of observations [38]. Our trajectories are ten rounds long. The claim that play becomes less rule-governed rests on exact matching and no learned component. The classifier was validated to 20% execution noise, which is two rounds in ten, and in two of the five models play sits about that far from the nearest canonical rule on average, so the read-out is working at the edge of the range it was validated on. The compositional claim uses labels the classifier supplies wherever the rules leave the answer open, is unevenly anchored across the four rules, and is the weaker of the two. Any reading of it as evidence that these models implement Tit-for-Tat would be unsupported by our data. Richer strategy vocabularies, including zero-determinant and extortionate play [26, 39] and mixture estimators over longer horizons [28, 37], are the natural extension.

Finally, the evolutionary baseline maps the stochasticity of a language model onto a symmetric per-action error rate, which is a clean theoretical object but a coarse abstraction: prompt sensitivity, end-game reasoning and the heterogeneity documented above are not well captured by symmetric flips. The baseline should be read as a reference distribution rather than as a fitted model, and the noise calibration reported in appendix C.6 is a first step rather than an estimate.

5. Conclusion

A positive rescaling of a payoff matrix is the rare manipulation whose predicted effect is not merely small but exactly zero. Five frontier language models were played across ten such rescalings, and every one of them moved. The movement is not a uniform bias a calibration could absorb: its shape differs by model and its size by prompt language, so most of it disappears from any analysis that averages models, and what survives the averaging depends on which models happen to be in the average. That is how a departure of this size can sit inside a literature that has not reported it. Read against an evolutionary baseline computed on the same matrices, the corpus moves the other way as the payoffs grow, which measures rather than explains the distance between what the game rewards and what these agents do.

What follows is a question about design rather than a verdict on any model. A cooperation rate measured on a language model is reported as though the matrix behind it were an arbitrary and therefore harmless choice. It is arbitrary, and it is not harmless. The same argument extends to every invariance the theory supplies, from shifts of the payoffs to relabelling of the actions, and each is an equally cheap test of whether an agent is playing the game it was shown.

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Ethics. This work did not involve human participants, animal subjects or fieldwork, and did not require ethical approval from a human subject or animal welfare committee; all experiments are computational simulations of interactions between large language model agents.

Data Accessibility. The corpus of game transcripts, the prompt templates, the trained strategy classifier and the complete analysis pipeline that produces every figure, table and quoted number in this article are available to the editors and reviewers now, at **[REPOSITORY URL, PRIVATE FOR REVIEW]**, with the transcripts released under **[DATA LICENCE]** and the analysis code under **[CODE LICENCE]**, and will be deposited in a public repository with a permanent DOI on acceptance.

Declaration of AI Use. Large language models are the *object* of study in this article: the corpus consists of transcripts of games played by the six models named in §2.3, collected through an automated pipeline. Generative AI was additionally used as a coding assistant for parts of the analysis and figure code, and for language editing of the manuscript. All analyses were designed, run and checked by the authors, every number reported here is produced by the archived pipeline and verified against the data by an automated script, and the authors take full responsibility for the content of this article.

Authors' Contributions. T.A.H.: conceptualisation, methodology, supervision, writing (original draft), writing (review and editing), funding acquisition. T.-K.H.: conceptualisation, software, formal analysis, visualisation, writing (original draft). D.-S.D.-M.: software, data curation, formal analysis, writing (original draft). T.-B.C., P.-H.L., H.-D.N., P.-Q.N.-L., M.-L.N.-V., H.-P.P., P.-H.P., T.-K.T., C.-N.T., H.T. and G.-T.T.-L.: investigation, data curation, validation, writing (review and editing). A.B.: methodology, resources, writing (review and editing). L.H.T.: conceptualisation, methodology, supervision, project administration, writing (review and editing). T.A.H., T.-K.H. and D.-S.D.-M. contributed equally to this work. All authors gave final approval for publication and agree to be held accountable for the work performed therein.

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